Warmup/Attendance: [Week 4: Wednesday 9/16/2026]

Full Name: | | SID: | |

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

Correlation

1. If the correlation coefficient for a given scatterplot is $r = -0.81$, which of the following must be true about the relationship between the explanatory and response variable?

a) The association has a linear form.
b) The association must be positive.
c) The association must be negative.
d) The association is weakened by outliers.

Simple Linear Model

2. In a study, volunteers kept a record of all the food they ate. The amount of protein (in grams per week) and the amount of fat (in grams per week) was then calculated for each person. Here is a summary of the entire data set:

Fat (g/wk)	Protein (g/wk)
mean = 668	mean = 662
sd = 65	sd = 70

Correlation(fat, protein) = 0.8

Calculate the equation of the least-squares regression line to predict protein in terms of fat.

3. A pediatric clinic models the relationship between a child's age in months (x) and their average weight in kilograms (y). Using data from thousands of healthy toddlers, they calculate the line of best fit:

predicted y = 0.4x + 7.2

Provide a verbal interpretation for the slope (0.4)